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Weiguo Wang; Tian Bai; Gaofei Xue; Hualong Sun; Xiaping Chen; Wenxi Guo ✉; Meidan Ye ✉ 



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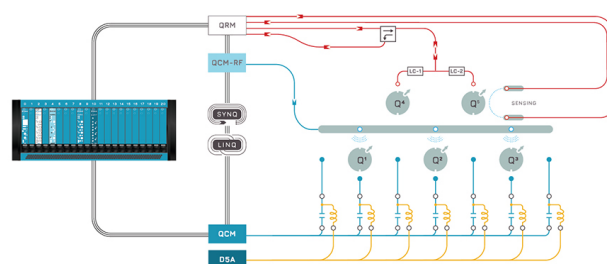


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Weiguo Wang,^{1,2} Tian Bai,^{1,2} Gaofei Xue,^{1,2} Hualong Sun,¹ Xiaping Chen,³ Wenxi Guo,^{1,2,a)} and Meidan Ye^{1,2,a)} 

AFFILIATIONS

¹Research Institute for Biomimetics and Soft Matter, Fujian Provincial Key Lab for Soft Functional Materials Research, Department of Physics, College of Physical Science and Technology, Xiamen University, Xiamen 361005, China

²Shenzhen Research Institute of Xiamen University, Shenzhen 518057, China

³Department of Physics, College of Physical Science and Technology, Xiamen University, Xiamen 361005, China

^{a)}Authors to whom correspondence should be addressed: wxguo@xmu.edu.cn and mdye@xmu.edu.cn

ABSTRACT

The quality of perovskite crystallization directly determines the performance of perovskite solar cells (PSCs). However, there is a lack of research on the bottom crystallization of perovskite films of PSCs. Here, by simply controlling the annealing temperature of perovskite films, CsPbIBr₂ films with different surface textures on both the top and bottom sides were obtained, respectively. It is found that dimethyl sulfoxide (DMSO), which is widely used to enhance the crystallization of perovskite films, cannot be completely removed under 160 °C, leading to the formation of snowflake-like textures (SFT) composed of countless pinholes on the upper and lower surfaces of the perovskite film and the existence of obvious cavities between the perovskite layer and the TiO₂ layer. This is mainly due to the dissolution of CsPbIBr₂ in the residual DMSO. On the contrary, CsPbIBr₂ films with compact and smooth surface textures (CST) on both the top and bottom sides can be achieved under 250 °C. For PSCs, the CST-based champion device (9.66%) exhibits higher efficiency and better stability than the SFT-based one (3.69%), primarily due to the high quality film with superior carrier separation and migration properties. The proposed understanding and accurate control of the perovskite crystal growth process are of great significance to the future development of PSCs.

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Perovskite solar cells (PSCs) have attracted extensive attention in recent decades. From 2009 to the present, the photoelectric conversion efficiency has been increased from 3.8% to exceeding 25%.¹ Because of its large absorption coefficient, low exciton binding energy, long carrier diffusion length, and low solution preparation cost,^{2,3} it has great development potential. However, the organic components, methylammonium (MA) and formamidinium (FA), in organic-inorganic hybrid perovskite are prone to decompose under heating conditions (>100 °C), which is inconsistent with the stability required by solar cells.⁴ It is found that Cs can effectively solve the problem of poor thermal stability by replacing organic components in perovskite to form CsPbI_xBr_{3-x}, which can maintain the crystal structure in a high-temperature environment (>400 °C).⁵

Inorganic perovskites usually adopt high boiling polar solvents to dissolve perovskite precursors, such as γ -butyrolactone (GBL), N, N'-dimethylformamide (DMF), dimethyl sulfoxide (DMSO), N-methyl-2-pyrrolidone (NMP), and their mixtures.⁶ Among them, DMSO has been proved to be a successful additive that can effectively improve the crystallization quality of perovskite. However, the

inherent high boiling point of DMSO makes it more difficult to evaporate during annealing.⁷ In order to obtain high-quality perovskite films, various methods have been developed to control the removal of solvents and the crystallization of perovskite. Yang *et al.* reported that DMSO and PbI₂ can form the DMSO-PbI₂ mesophase and effectively slow down the crystallization.⁸ Xiao *et al.* found that thermal annealing can effectively increase the grain size and film coverage of perovskite films and improve the efficiency of PSCs.⁹ Kim *et al.* adopted the high temperature and short-time annealing treatment to make the solvent evaporate rapidly, resulting in uniform grains and large-size grain domains of perovskite films in a large area.¹⁰

Although various methods have been developed to control the crystallization process of perovskites, there is still a lack of understanding of the crystallization process of perovskite grains from liquid to solid states. The previous studies mainly focused on the grain growth along the plane direction and on the upper surface of perovskites (top side), while the research on the lower surface of perovskite films (bottom side) is rarely studied.¹¹ This leads to the one-sided understanding of the perovskite grain growth and cannot fully and

accurately understand the overall perovskite crystallization process, which is quite unfavorable for the future industrialization of PSCs.

Here, CsPbIBr₂ films with different surface textures [i.e., snowflower-like textures (SFT) and compact and smooth textures (CST)] were obtained by controlling the annealing temperature, respectively. Their crystallization on the top and bottom sides of perovskite films was carefully discussed. It indicated that DMSO cannot be completely removed by the low-temperature (160 °C) annealing, and the residual DMSO will etch perovskites, resulting in numerous pinholes forming snowflower-like textures on both the upper and lower surfaces, and causing obvious cavities between the perovskite layer and the TiO₂ layer. Thrilling, simply raising the temperature (250 °C) can effectively promote the sufficient removal of DMSO and realize the better crystallization of perovskite films. Finally, CsPbIBr₂ films with different surface textures were separately assembled into solar cell devices, and the adverse effect of notorious snowflakes on the device efficiency was fully explored. The CST-based PSCs showed much superior performance than the SFT-based PSCs.

As shown in Fig. 1(a), the perovskite films with CST were obtained at 250 °C, while the perovskite films covered with SFT were obtained at 160 °C. The detailed process can be found in the [supplementary material](#). The quality of these two kinds of films can be clearly

distinguished under the scanning electron microscope (SEM). As indicated in Figs. 1(b), 1(c), and S1, the surface of the SFT CsPbIBr₂ film presents well-defined snowflower-like textures at low magnification. By gradually enlarging the top view of surface area, it is found that there are countless pinholes in the snowflower area [Figs. 1(d) and 1(e)], which will lead to serious non-radiative recombination of carriers.^{4,12} The reason for the formation of these pinholes may be that DMSO has a high boiling point (189 °C) and is difficult to remove at low temperature, and thus, the solvent may remain under the perovskite film during the perovskite annealing.¹³ Consequently, if the solvent does not evaporate in time and a large amount of DMSO rises from the bottom, it is quite possible to redissolve the perovskite film and form a large number of pinholes on the surface of the perovskite films.^{14,15} Figures 1(f)–1(i), and S2 present the top view of CST films. It shows high-quality crystalline perovskite films without pinholes from low magnification to high magnification, indicating that high temperature can effectively reduce the pinholes on the surface of perovskite film and improve the film quality. This may be related to the rapid removal of solvent at high temperature, but it does not mean that higher temperature is more conducive to the perovskite crystallization. As shown in Fig. S3(a), perovskite films tend to form smaller grains at a high temperature of 300 °C, suggesting that there are more grain

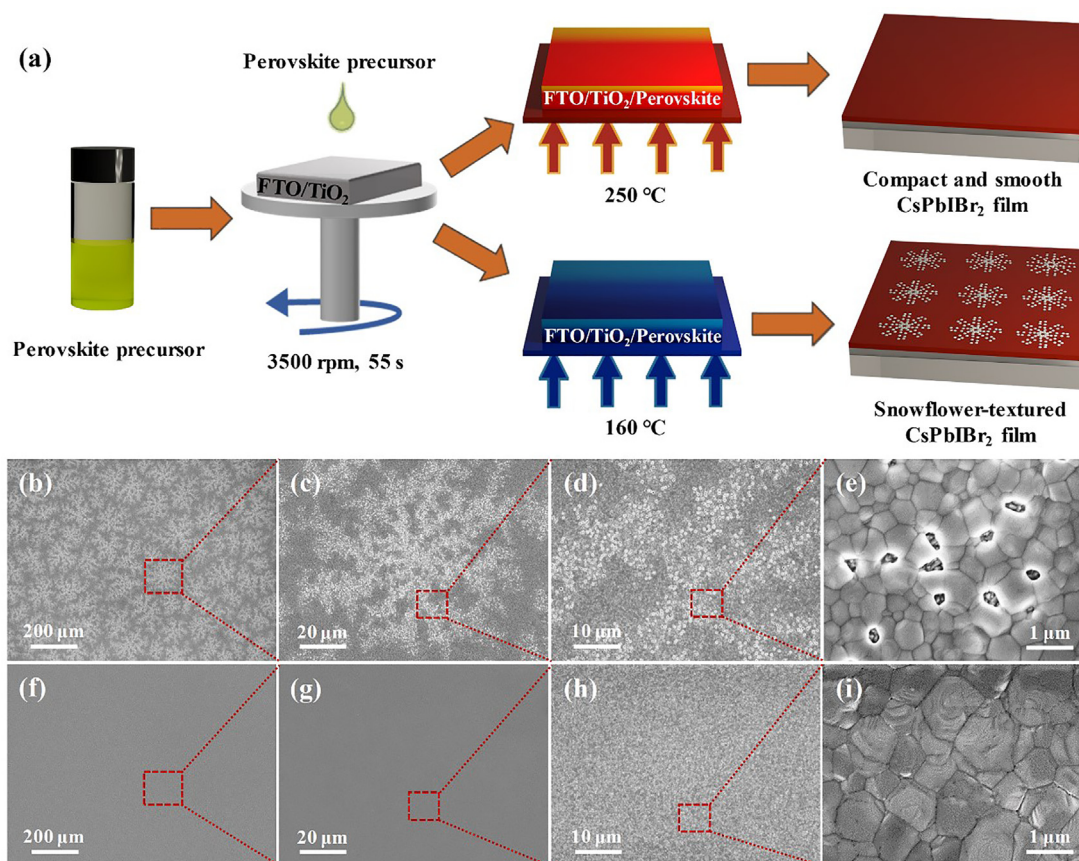


FIG. 1. (a) Schematic diagrams of the preparation of CsPbIBr₂ films with different surface textures. SEM images (b)–(i) of different CsPbIBr₂ films under different magnification: snowflower-like surface textures [SFT; (b)–(e)], and compact and smooth surface textures [CST; (f)–(i)], respectively.

boundaries for more non-radiative recombination, which is unfavorable for the improvement of power conversion efficiency [PCE, Fig. S3(b)]. The polarizing microscope pictures (Fig. S4) of CST and SFT films further indicate their obviously different surface textures.

Furthermore, Figs. 2(a)–2(c) show the behavior diagram of DMSO solvent during the annealing process of perovskites. It is difficult for the DMSO solvent molecules to quickly evaporate, and thus, some of them will stay under the perovskite film during the 160 °C-annealing process.¹⁶ Figure S5 shows the Fourier transform infrared (FTIR) spectra of CST and SFT annealed at different times. The stretching vibration at 1056 cm^{-1} is caused by S=O in DMSO.¹⁷ It suggests that DMSO remains in SFT after annealing for 10 min, while it takes only 7 min for CST to remove DMSO completely. It may lead to the dissolution of perovskite by DMSO, resulting in the formation of snowflower-like surface textures and even some obvious cavities between the perovskite layer and the TiO_2 -coated substrate [Fig. 2(d)]. Under a high temperature of 250 °C, DMSO evaporates more fully and faster, and perovskite grains are not dissolved. Therefore, the resultant perovskite film maintains compact and smooth surface textures, and the perovskite layer and the TiO_2 -coated substrate combine closely without significant gaps [Fig. 2(e)].

In addition to the upper surface textures of perovskite films, their lower surface textures are also very essential to the interfacial contact between the electron transporting layer and the perovskite layer.

Hence, it is necessary to peel the perovskite layer from the substrate to observe the lower surface texture of CsPbIBr_2 film, and the detailed process can be found in the [supplementary material](#). As shown in Figs. S6 and S7, the surface textures of perovskite films can be completely retained.¹⁸ Figures 2(f)–2(h) demonstrate that the lower surface textures of the SFT film are loose and rough with many snowflower-like defects, while the CST film still has compact and smooth surface textures of its lower side [Fig. 2(i)]. It can be seen that high temperature also can help the grain on the lower surface of perovskite connect to each other more closely, which is conducive to the carrier transport in devices.¹⁹ As shown in Fig. S8, the difference of upper and lower surface textures between SFT and CST even can be visible from their optical pictures. Moreover, the SFT film has a larger film roughness (37.5 nm) as compared to that (15.4 nm) of the CST film measured by the atomic force microscope (Fig. S9). Except for the deposition of perovskite films on the TiO_2 -covered FTO substrate, similar surface textures also can exist on the SnO_2 -covered FTO substrate and even the pure FTO substrate (Fig. S10), indicating the universality of temperature effect on the perovskite crystallization.

To explore the difference between the CST and SFT perovskite films, more characterization was applied to these two films. As shown in Fig. S11(a), these two perovskite films have a similar energy bandgap of about 2.07 eV,²⁰ and the CST film has a slightly higher light absorption capacity than that of the SFT film. Figure S11(b) exhibits the x-ray

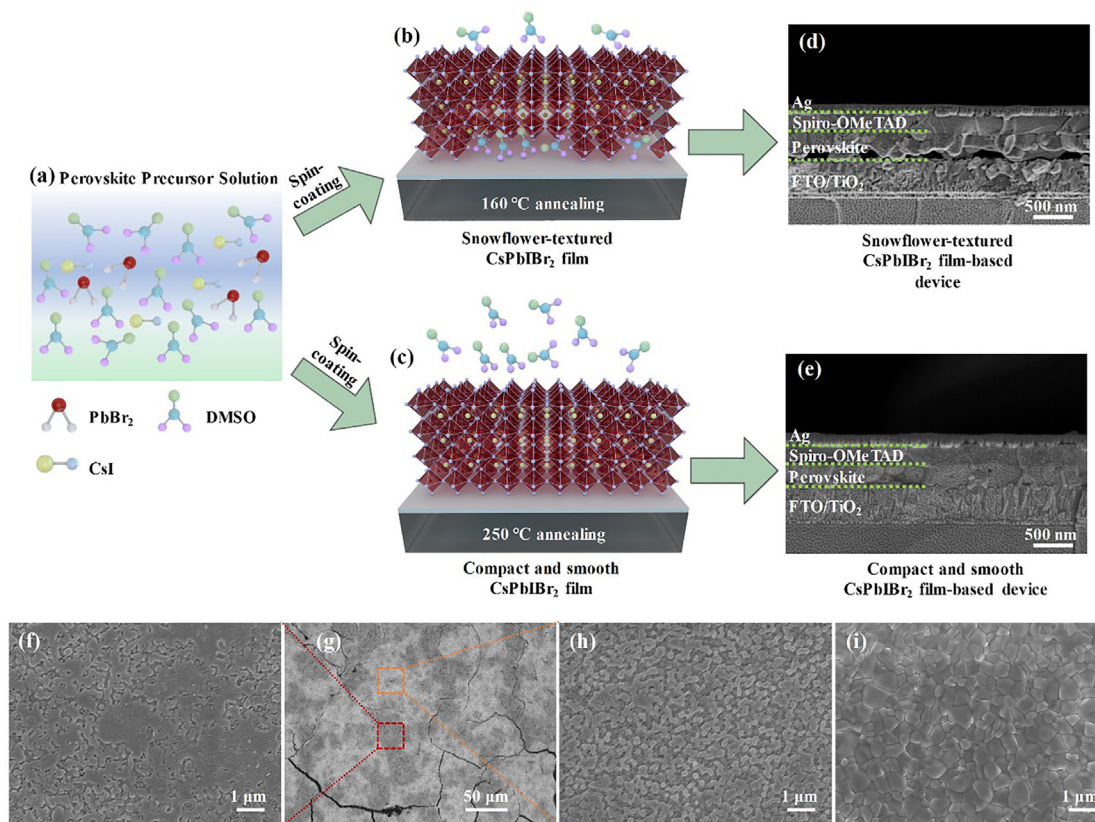


FIG. 2. Schematic diagrams of the perovskite growth processes (a)–(c) and SEM images of the cross-sectional views of (d) SFT-, (e) CST-based PSC devices, respectively. SEM images of the lower surface (bottom side) textures of (f)–(h) SFT and (i) CST CsPbIBr_2 films, respectively.

diffraction (XRD) patterns of these two films with typical diffraction peaks of CsPbIBr₂, including $2\theta = 14.6^\circ$, 20.8° , and 29.7° for the (100), (110), and (200) crystal planes of perovskite, respectively.²¹ The diffraction peak intensity of CST is relatively higher than that of SFT, indicating that CST has stronger crystallinity. Figure S11(c) is the photoluminescence steady-state emission spectra of these two films. The peak intensity of CST is lower than that of SFT, suggesting that CST has better charge extraction efficiency.²² From the full spectrum scanning of the x-ray photoelectron spectroscopy (XPS) revealed in Fig. S11(d), there is no difference in the composition between CST and SFT films.²³ Their chemical states can be studied by their XPS core spectra of Cs 3d, Pb 4f, I 3d, and Br 3d in Fig. S12.²⁴ The peak position of CST moves toward higher binding energy as compared to that of SFT, which may be due to the more stable lattice structure of CST.²⁵

In order to explore the influence of CST and SFT films on the device performance of PSCs, they were assembled into PSCs with a device structure of FTO/TiO₂/perovskites/Spiro-OMeTAD/Ag, respectively. All devices were tested under one standard sunlight. As shown in Fig. 3(a), the PCE of the champion CST-based device is 9.66% with an open-circuit voltage (V_{OC}) of 1.22 V, a short-circuit

current density (J_{SC}) of 10.87 mA cm^{-2} , and a filling factor (FF) of 72.95%. In contrast, the best PCE of the SFT-based device decreases sharply with only 3.69% ($V_{OC} = 1.17 \text{ V}$, $J_{SC} = 6.36 \text{ mA cm}^{-2}$, FF = 49.50%). Similar results can be observed in the SnO₂-based PSC devices (Fig. S13). The pinholes on the upper and lower surface of the SFT CsPbIBr₂ film and the obvious gaps between the perovskite layer and the TiO₂ layer bring great inconvenience to the carrier transmission and seriously reduce the carrier life in the device. For the data reliability, the PCE, J_{SC} , V_{OC} , and FF values of 40 devices based on the CST and SFT films were recorded and plotted, respectively. As shown in Figs. 3(b)–3(e), it can be concluded that the SFT-based PSCs are far inferior to those of the CST-based ones in every parameter. Moreover, according to their respective incident photon-to-electron conversion efficiency (IPCE) tests [Fig. 3(f)], the optical response of SFT to each wavelength of light is worse than that of CST, especially for the light in the range of 400–600 nm, and their integral J_{SC} values are close to those from the J – V measurement.

Furthermore, Fig. 4(a) presents the dark J – V curves of two types of PSCs, indicating that the dark current under high bias voltages and the leakage current under low bias voltages of the CST device are both

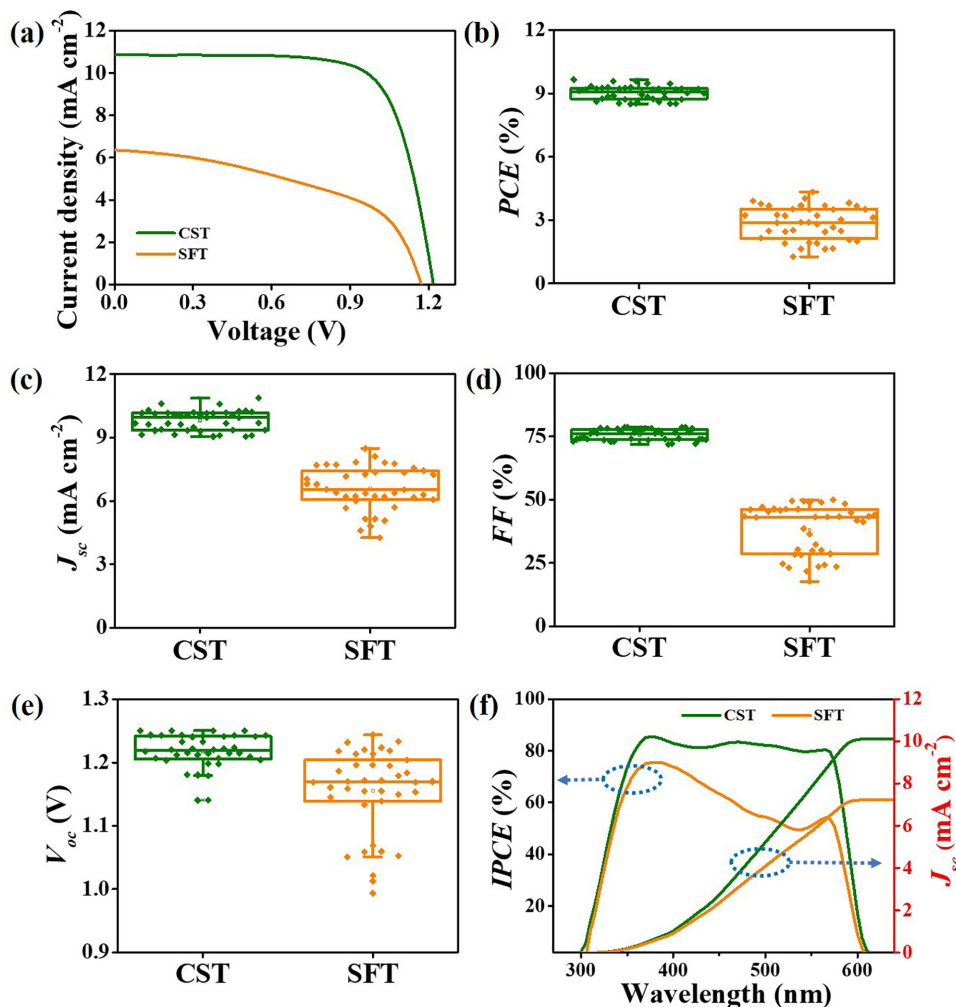


FIG. 3. Device performance: (a) J – V curves of the CST- and SFT-based PSCs. (b) PCE, (c) J_{SC} , (d) FF, and (e) V_{OC} plots for 40 devices based on the CST and SFT, respectively. (f) IPCE curves of the CST- and SFT-based PSCs.

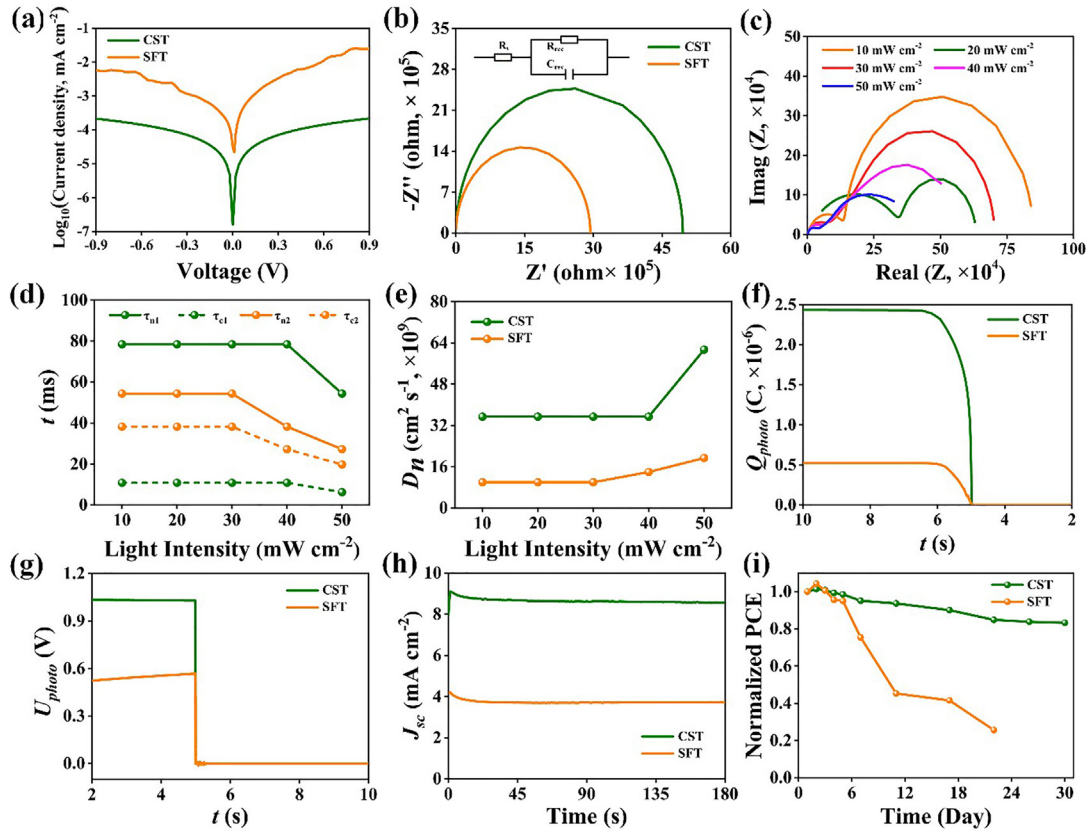


FIG. 4. Device performance: (a) J - V curves in the dark state, (b) EIS spectra; (c) IMVS (Nyquist plot) of CST-based PSCs; light intensity dependence of (d) carrier lifetime and carrier transmission time, (e) carrier diffusion coefficients; (f) charge extracted within 5 s, and (g) open circuit voltage under 50 mW cm^{-2} light irradiation of the CST- and SFT-based PSCs, respectively. (h) The stabilized maximum power output tests; (i) change in PCE values of the CST- and SFT-based PSCs stored in the 20%–40% humidity ambient environment.

lower than that of the SFT device. It means that the former may have greater V_{OC} .^{26,27} The electrochemical impedance spectrum (EIS) is implemented to study the carrier transmission characteristics of PSCs. The bias voltage was applied under dark conditions. The obtained Nyquist plot is shown in Fig. 4(b), and the fitted equivalent circuit diagram is also included. The high-frequency region represents the resistance (R_s) in the process of carrier transmission, which is usually considered to be related to the charge transmission from anode to cathode. The low-frequency region represents the composite resistance (R_{ec}) and chemical capacitance (C_{ec}) at the interface between the perovskite layer and the electron/hole transport layers. The CST-based device has a smaller R_s value ($0.95 \text{ k}\Omega$) than that of the SFT based devices ($1.80 \text{ k}\Omega$), indicating that it has better carrier transmission performance, while a larger R_{ec} value suggests that CST ($4.95 \times 10^3 \text{ k}\Omega$) has a higher carrier composite barrier among different functional layers than SFT ($2.93 \times 10^3 \text{ k}\Omega$), which is conducive to the carrier injection and transfer of photovoltaic devices.²⁸

The behavior of carriers is worthy of further study. Intensity modulated voltage spectroscopy (IMVS) and intensity modulated photocurrent spectroscopy (IMPS) were applied to study the transmission and lifetime of carriers in PSCs. The Nyquist plots of IMVS and IMPS are shown in Figs. 4(c) and S14, and then according to the formula^{29,30}

$$\tau_n = \frac{1}{2\pi f_{\min}}, \quad (1)$$

$$\tau_c = \frac{1}{2\pi f_{\min}}, \quad (2)$$

where $f_{\min}$ refers to the frequency of the lowest point of the Nyquist diagram, the carrier life (τ_n) and carrier transmission time (τ_c) can be easily obtained. Figure 4(d) shows the carrier lifetime and transmission time of CST and SFT under different light intensities. The CST-based device always displays a longer carrier lifetime and a shorter transmission time than those of the SFT-based device, suggesting that CST has better electrical performance and lower carrier recombination than SFT.³¹ Moreover, their diffusion coefficient (D_n) can be determined by the formula³²

$$D_n = \frac{d^2}{2.35\tau_c}, \quad (3)$$

where d is the film thickness, which can be obtained from the cross section of the devices in Fig. 2. Figure 4(e) displays the diffusion coefficient of CST and SFT-based devices under different light intensities, suggesting that the electronic transmission of CST-based device is faster than that of the SFT-based device.³³ The parameters derived from the electrochemical test are given in Table S1.

Thus, the CST-based device has a higher J_{SC} value than that of the SFT-based device, which is due to its longer carrier life and faster carrier transmission speed, agreeing well with the previous characterization results. The poor carrier transport of SFT is directly related to the defects in perovskite and the poor contact between the perovskite layer and the electron transport layer.

Figure 4(f) presents the charge extracted from the device after the device changes to the short-circuit state when the light is turned off, which can also reflect that the CST-based device has better electrochemical performance.³⁴ In order to further observe the change of open-circuit voltage, the devices were also placed under the light of 50 mW cm⁻² for 5 s. As shown in Fig. 4(g), the CST-based device keeps in a stable open-circuit voltage with a higher value during illumination, while the open-circuit voltage of SFT has been climbing steadily, implying that the voltage output of SFT-based device remains in an unstable state.³⁰ When a standard light was applied, the J_{SC} value of the device was detected to observe the stability of the continuous output. It is observed in Fig. 4(h) that during the output process in 180 s, the J_{SC} value of the CST-based device is maintained at 9 mA cm⁻², while that of the SFT-based device is only maintained at 4 mA cm⁻², implying that CST has better output stability and higher output currents. Figure 4(i) exhibits the stability of devices without encapsulation stored in an ambient environment with a humidity of 20%–40%. The CST-based device can still maintain more than 80% of the original efficiency after 30 days, while the efficiency of the SFT-based device reduces to 25% of the original value after 22 days. The stronger humidity resistance is due to the better crystallization of CST perovskite films. SFT has many pinholes in perovskite films, and thus, it is easily eroded by water.

In summary, CsPbIBr₂ films with different surface textures were prepared by controlling the annealing temperature. It is found that the residual DMSO in the perovskite film at 160 °C will lead to the formation of snowflake-like surface textures on the top and bottom sides of the perovskite layer, resulting in the quality decline of films. Thrillingly, high temperature can effectively remove the residual DMSO and obtain higher quality perovskite films. SFT does harm to carrier life and transmission, as well as to the electrical properties of the resultant PSC, and damages the device stability. While CST shows superior electrical properties and stability. The discussion and the powerful optimization of perovskite crystallization on both the top and bottom sides by the temperature strategy are of great significance for the industrialization of PSCs in the future.

See the [supplementary material](#) for experimental section, morphological/elemental characterization of perovskite films and performance tests of devices.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its [supplementary material](#).

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